Absence of Secular Generalized-Energy Drift

A backward-error argument for Gauss–Legendre and fully extended ABBA methods

Precise theorem, proof, scope, and implementation limitations for GC2D

1 Purpose and precise claim

This note proves a conditional near-conservation result for fixed-step symplectic discretizations of the time-dependent guiding-centre problem. The central conclusion is not exact conservation and is not a statement for infinite time. Under the analytic backward-error hypotheses stated below, a symplectic method of order p satisfies

$$|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \le Ch^p \qquad \text{for} \qquad t_n = nh \le \exp(c/h), \tag{1}$$

after a harmless reduction of the positive constant c. The constants are independent of h and n, but they depend on the Hamiltonian, the complex analyticity domain, and the compact set containing the numerical orbit.

Equation (1) excludes a secular contribution of the form $t_n h^p$ throughout the stated interval. It does not imply that the energy error is exactly periodic, that its average is zero, or that the same bound holds as $n \to \infty$ with h fixed.

The result applies to the autonomous extended Hamiltonian

$$\mathcal{K}(x, y, t, k) = H(t, x, y) + k, \tag{2}$$

not to the explicitly time-dependent physical value $H(t, x, y)$ alone.

2 Physical and extended Hamiltonian systems

For one guiding-centre particle, let

$$z = (x, y)^T, \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The physical system is

$$\dot{z} = J_0 \nabla_z H(t, z) = \begin{pmatrix} -H_y(t, x, y) \\ H_x(t, x, y) \end{pmatrix}. \tag{3}$$

Along an exact physical trajectory,

$$\begin{aligned} \frac{d}{dt} H(t, z(t)) &= H_t(t, z) + \nabla_z H(t, z)^T \dot{z} \\ &= H_t(t, z) + \nabla_z H(t, z)^T J_0 \nabla_z H(t, z) = H_t(t, z). \end{aligned} \tag{4}$$

The last equality uses the skew-symmetry of J_0. Consequently, H is not an invariant when $H_t \neq 0$.

Introduce the extended state and constant Poisson matrix

$$Z = (x, y, t, k)^T, \qquad \mathbb{J} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \tag{5}$$

With $\mathcal{K}$ from (2), the autonomous extended equations are

$$\dot{Z} = \mathbb{J}\nabla_Z \mathcal{K}(Z) = \begin{pmatrix} -H_y \\ H_x \\ 1 \\ -H_t \end{pmatrix}. \tag{6}$$

They reproduce the physical system, advance time at unit speed, and evolve the conjugate momentum according to $\dot{k} = -H_t$. Moreover,

$$\frac{d}{ds}\mathcal{K}(Z(s)) = \nabla_Z \mathcal{K}(Z)^T \mathbb{J} \nabla_Z \mathcal{K}(Z) = 0. \tag{7}$$

Thus $\mathcal{K} = H + k$ is the invariant whose numerical drift is meaningful.

Since $\mathbb{J}^{-1} = -\mathbb{J}$, the extended symplectic-form matrix is

$$\Omega = \mathbb{J}^{-1} = -\mathbb{J}. \tag{8}$$

A differentiable numerical map Φ_h is symplectic when

$$D\Phi_h(Z)^T \Omega D\Phi_h(Z) = \Omega. \tag{9}$$

3 Hypotheses for exponentially long near-conservation

The exponential time scale requires stronger assumptions than finite-interval convergence.

Assumption 3.1 (Analytic Hamiltonian)**.** There is a compact simply connected real domain $D \subset \mathbb{R}^4$ and a complex neighbourhood D_ρ of uniform width $\rho > 0$ on which $\mathcal{K}$ is analytic and bounded together with the derivatives required below.

Assumption 3.2 (Ideal numerical map)**.** For fixed h, the one-step map Φ_h is analytic on D_ρ, symplectic, and of order p:

$$\Phi_h(Z) = \varphi_h^{\mathcal{K}}(Z) + O(h^{p+1}), \tag{10}$$

uniformly on the relevant domain. Every implicit stage or projection equation is solved at its exact local root and arithmetic is exact.

Assumption 3.3 (Confined orbit and small fixed step)**.** The fixed step h is sufficiently small relative to the analytic scales of the problem, and the numerical orbit $Z_j = \Phi_h^j(Z_0)$ remains in a compact subset of D for all indices considered.

The qualification "sufficiently small" is problem dependent. For a Hamiltonian containing a fast frequency ω, the constants deteriorate as ω increases; resolving that scale, rather than merely requiring a small dimensional value of h, is essential.

4 Modified Hamiltonian

Backward-error analysis seeks a differential equation whose exact time-h flow reproduces the numerical map to a much higher accuracy than the original order estimate. Formally, the modified vector field has an expansion

$$F_h = F + h^p F_p + h^{p+1} F_{p+1} + \cdots, \qquad F = \mathbb{J}\nabla\mathcal{K}. \tag{11}$$

Proposition 4.1 (Hamiltonian character of the modified equation)**.** *Under Assumption 3.2, every coefficient of the formal modified vector field is locally Hamiltonian. On the simply connected domain D, one may therefore write*

$$F_h = \mathbb{J}\nabla\widetilde{\mathcal{K}}_h, \qquad \widetilde{\mathcal{K}}_h = \mathcal{K} + h^p\mathcal{K}_p + h^{p+1}\mathcal{K}_{p+1} + \cdots . \tag{12}$$

If the method is also symmetric,

$$\Phi_{-h} = \Phi_h^{-1}, \tag{13}$$

then the modified Hamiltonian may be chosen with the parity expansion

$$\widetilde{\mathcal{K}}_h = \mathcal{K} + h^p\mathcal{K}_p + h^{p+2}\mathcal{K}_{p+2} + h^{p+4}\mathcal{K}_{p+4} + \cdots . \tag{14}$$

Proof. The formal logarithm of a near-identity one-step map defines the modified vector field. Symplecticity of Φ_h implies, order by order, that its formal logarithm is a symplectic vector field. On a simply connected canonical domain, a symplectic vector field is Hamiltonian, which gives (12). The order-p consistency in (10) removes all corrections below h^p.

For a symmetric method, replacing h by $-h$ produces the inverse map. Taking the formal logarithm shows that the modified vector field is an even function of h. Hence the terms after h^p advance in increments of two, as stated in (14). Symmetry supplies this parity, but symplecticity is the property that makes the modified vector field Hamiltonian. □

The formal series generally diverges and cannot itself be declared an exactly conserved numerical energy. Analyticity permits optimal truncation before the coefficients begin to grow.

Lemma 4.2 (Exponentially accurate truncated modified Hamiltonian)**.** *Under Assumptions 3.1–3.3, there are positive constants C_0, C_1, c, and h_0 such that, for $0 < h < h_0$, the formal series can be truncated at an index $N(h) = O(h^{-1})$ to produce $\widetilde{\mathcal{K}}_h^{[N]}$ satisfying*

$$\sup_{Z\in D}\left|\widetilde{\mathcal{K}}_h^{[N]}(Z) - \mathcal{K}(Z)\right| \le C_0 h^p, \tag{15}$$

$$\left|\widetilde{\mathcal{K}}_h^{[N]}(\Phi_h(Z)) - \widetilde{\mathcal{K}}_h^{[N]}(Z)\right| \le C_1 h\exp(-c/h) \tag{16}$$

whenever the step and the connecting modified trajectory remain in the domain used for the estimates.

Proof sketch. Analyticity on the uniform complex neighbourhood gives Cauchy estimates for successive derivatives. Applied to the recursive construction of the modified equation, these estimates bound its coefficients by factorially growing quantities. Truncating near the smallest term, at $N(h) \asymp c/h$, converts the algebraic remainder into an exponentially small one. Because Proposition 4.1 makes the truncated modified equation Hamiltonian, its exact flow conserves $\widetilde{\mathcal{K}}_h^{[N]}$. The exponentially small difference between that flow and Φ_h then gives (16); consistency gives (15). This optimal-truncation result is the standard analytic input of Hamiltonian backward-error analysis. □

5 No-secular-drift theorem

Theorem 5.1 (Generalized-energy near-conservation)**.** *Under Assumptions 3.1–3.3, there are positive constants C, c, and h_0 such that the numerical orbit satisfies*

$$|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \le Ch^p + Ct_n\exp(-c/h), \qquad t_n = nh, \tag{17}$$

for every index for which the domain assumptions remain valid. Consequently, after possibly reducing c, there is a constant C' such that

$$\boxed{|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \le C'h^p \qquad \textit{when} \qquad 0 \le t_n \le \exp\left(\frac{c}{2h}\right).} \tag{18}$$

Proof. Let $\widetilde{\mathcal{K}}_h^{[N]}$ be supplied by Lemma 4.2. Telescoping its one-step defect along the numerical orbit gives

$$\begin{aligned}\left|\widetilde{\mathcal{K}}_h^{[N]}(Z_n) - \widetilde{\mathcal{K}}_h^{[N]}(Z_0)\right| &\le \sum_{j=0}^{n-1} \left|\widetilde{\mathcal{K}}_h^{[N]}(Z_{j+1}) - \widetilde{\mathcal{K}}_h^{[N]}(Z_j)\right| \\ &\le nC_1 h \exp(-c/h) = C_1 t_n \exp(-c/h). \end{aligned} \tag{19}$$

Insert and subtract the truncated modified Hamiltonian at both endpoints:

$$\begin{aligned}|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \le &\left|\mathcal{K}(Z_n) - \widetilde{\mathcal{K}}_h^{[N]}(Z_n)\right| \\ &+ \left|\widetilde{\mathcal{K}}_h^{[N]}(Z_n) - \widetilde{\mathcal{K}}_h^{[N]}(Z_0)\right| + \left|\widetilde{\mathcal{K}}_h^{[N]}(Z_0) - \mathcal{K}(Z_0)\right|. \end{aligned} \tag{20}$$

Equations (15) and (19) yield

$$|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \le 2C_0 h^p + C_1 t_n \exp(-c/h),$$

which is (17) after renaming constants.

If $t_n \le \exp(c/(2h))$, the accumulated exponential term is bounded by $C_1 \exp(-c/(2h))$. For every fixed p, this is smaller than a constant multiple of h^p when h is sufficiently small. Absorbing it into the first term proves (18). □

Remark 5.2 (Meaning of "no secular drift"). Theorem 5.1 rules out an energy error growing like $t_n h^p$ on the exponentially long interval. It proves confinement to an $O(h^p)$ band, not literal oscillation about zero. A plotted sign change or small fitted slope is useful evidence, but it is not part of the theorem.

Remark 5.3 (Not an infinite-time theorem). For fixed h, the upper time in (18) is finite. The theorem therefore does not exclude later effects caused by resonances, loss of confinement, exponentially small remainders, or arithmetic errors.

6 Application to the GC2D integrators

6.1 Two-stage Gauss–Legendre

Let $A = (a_{ij})$, $b = (b_i)$, and $c = (c_i)$ be the two-stage Gauss–Legendre coefficients. Applying this Runge–Kutta method to (6) gives the stage times

$$T_i = t_n + h \sum_j a_{ij} = t_n + c_i h, \tag{21}$$

because $A\mathbf{1} = c$. The physical stages solve

$$Z_i^{\mathrm{phys}} = z_n + h \sum_j a_{ij} J_0 \nabla_z H(T_j, Z_j^{\mathrm{phys}}), \tag{22}$$

and the accepted physical and momentum updates are

$$z_{n+1} = z_n + h \sum_i b_i J_0 \nabla_z H(T_i, Z_i^{\mathrm{phys}}), \tag{23}$$

$$k_{n+1} = k_n - h \sum_i b_i H_t(T_i, Z_i^{\mathrm{phys}}). \tag{24}$$

The internal k stages need not be solved because neither the physical field nor $-H_t$ depends on k. Thus the triangular implementation of (24) is algebraically the Gauss method on the full extended

system. This is the extended map represented by `GaussLegendre4()` with an `EnergyObserver` in the project implementation.

Gauss coefficients satisfy the symplectic Runge–Kutta identities

$$b_i a_{ij} + b_j a_{ji} = b_i b_j, \qquad i, j = 1, 2. \tag{25}$$

The method is symmetric and has order four.

Corollary 6.1 (Ideal Gauss–Legendre energy band)**.** *If Assumptions 3.1 and 3.3 hold and the Gauss stage equations are solved exactly, then*

$$|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| = O(h^4) \tag{26}$$

through the exponentially long interval of Theorem 5.1.

6.2 Fully extended implicit ABBA

The fully extended ABBA construction duplicates the complete canonical state $Z = (x, y, t, k)$, applies a composition of Hamiltonian shears on the doubled space, and returns to the diagonal through an implicit symmetric projection. The exact projected map is symplectic under the following local hypotheses:

- every doubled shear is evaluated exactly;
- the projection root is locally unique;
- the residual Jacobian with respect to the multiplier is nonsingular;
- the projection equation is solved exactly.

The physical ABBA2 projection proof and its local hypotheses are developed in the model note `ABBA2_implicit.tex`, located under `docs/models/abba2-implicit/`. The fully extended construction applies the same symmetric-projection mechanism to the complete canonical state. Palindromic compositions of the projected second-order map remain symplectic, and the single-projection fourth-order variant applies the same mechanism around the complete symplectic base composition.

Corollary 6.2 (Ideal fully extended ABBA energy bands)**.** *Under the local exact-projection hypotheses above and Assumptions 3.1 and 3.3, the fully extended implicit ABBA methods satisfy the bounds in Table 1.*

Table 1: Ideal generalized-energy bands from symplectic backward-error analysis.

Method	Order p	Energy band
Two-stage Gauss–Legendre	4	$O(h^4)$
`ABBA2Implicit`, fully extended	2	$O(h^2)$
`ABBA4Implicit`, per-map projection, fully extended	4	$O(h^4)$
`ABBA4Implicit`, exterior projection, fully extended	4	$O(h^4)$
`ABBA6Implicit`, fully extended	6	$O(h^6)$

The statement does not cover `ABBA2Midpoint`: arithmetic averaging of the final copies is not a symplectic projection in general. Nor does it automatically cover an energy diagnostic obtained by advancing or reconstructing k after a physical-only projected map. For a theorem about $\mathcal{K}$, the accepted map must be symplectic in the complete canonical extended space, not merely area preserving in (x, y).

7 Finite regularity of the runtime potential

The analytic hypothesis is substantive for the current implementation. A tensor-product spline of degree q is generically only C^{q-1} across simple internal knots. It is polynomial, hence analytic, inside one open spline cell, but a trajectory crossing multiple knots does not remain in one common analytic expression. The HDF5 evaluator periodically wraps coordinates and extends the sampled values across the cell seam. It therefore introduces no separate clipping discontinuity; crossings of the periodic seam have the same finite-smoothness limitation as crossings of ordinary spline knots.

Consequently, Theorem 5.1 is a rigorous theorem for an analytic idealization, but its exponential time scale is not a global theorem for the periodic-spline runtime representation. Finite-smoothness backward-error analysis can still construct a modified Hamiltonian through a finite order. If a truncation at order $N > p$ gives a one-step modified-energy defect $O(h^{N+1})$, the same telescoping argument yields schematically

$$|\mathcal{K}(Z_n) - \mathcal{K}(Z_0)| \leq Ch^p + Ct_n h^N. \tag{27}$$

This keeps an $O(h^p)$ band only on an algebraic time scale such as

$$t_n \leq ch^{-(N-p)}, \tag{28}$$

and only when the available differentiability and uniform derivative bounds justify the chosen N.

There are two clean ways to strengthen the analytic claim:

1. use a globally analytic spatial representation, such as a finite Fourier representation on a periodic domain; or
2. prove a finite-regularity theorem with an explicit N supported by the chosen spline degree, including ordinary knots and the periodic seam.

8 Finite nonlinear solves and floating-point arithmetic

Let Φ_h denote the exact-root symplectic map and let the implemented map be

$$\widehat{\Phi}_h(Z) = \Phi_h(Z) + d_h(Z), \qquad \|d_h(Z)\| \leq \eta_h. \tag{29}$$

Data-dependent Newton or Broyden stopping generally makes $\widehat{\Phi}_h$ neither exactly symplectic nor exactly symmetric. If the truncated modified Hamiltonian is Lipschitz on the numerical domain, the additional one-step change caused by d_h is at most $C\eta_h$. Repeating the proof of Theorem 5.1 gives the worst-case estimate

$$\left|\mathcal{K}(\widehat{Z}_n) - \mathcal{K}(\widehat{Z}_0)\right| \leq Ch^p + Ct_n \exp(-c/h) + C\frac{t_n}{h}\eta_h. \tag{30}$$

The final term is potentially secular. If the target integration time is T, retaining an $O(h^p)$ energy band in this worst-case estimate requires

$$\eta_h \ll \frac{h^{p+1}}{T}. \tag{31}$$

The nonlinear residual tolerance is not itself η_h. If the root Jacobian is uniformly invertible and its inverse is bounded by M, a local perturbation argument gives a relation of the form

$$\eta_h \leq CM\tau_h, \tag{32}$$

where τ_h bounds the accepted residual and the constant also depends on the reconstruction map. This conditioning factor must be included in any rigorous implementation-level estimate.

Round-off adds another perturbation. A random-walk growth proportional to $\sqrt{n}$ is often observed, but it is a statistical model, not a deterministic guarantee. An adversarial deterministic bound must allow growth proportional to n.

9 What is proved, observed, and not claimed

- **Proved for the ideal analytic map:** Gauss–Legendre and the fully extended implicit ABBA variants have an $O(h^p)$ generalized-energy band over exponentially long, but finite, times.
- **Proved algebraically for the exact physical flow:** $H(t, z)$ is not generally conserved, whereas $\mathcal{K} = H + k$ is conserved.
- **Not proved by a bounded-looking graph:** sign changes, a small final-to-peak ratio, and a small fitted slope support the interpretation but do not establish the long-time theorem.
- **Not covered by the exponential theorem as currently represented:** spline-knot and periodic-seam crossings, finite nonlinear stopping, and floating-point arithmetic.
- **Not claimed:** exact conservation of $\mathcal{K}$, a bound for all n, or a bounded physical difference $H(t_n, z_n) - H(0, z_0)$.

If exact step-by-step generalized-energy conservation is the primary goal, an energy-preserving discrete-gradient method gives a different algebraic result,

$$\mathcal{K}(Z_{n+1}) = \mathcal{K}(Z_n), \tag{33}$$

at the exact nonlinear root. Such a method is not generally symplectic. This is stronger in energy and weaker in symplectic structure than the near- conservation result proved here.

10 Conclusion

The absence of secular generalized-energy drift for symplectic GC2D integrators is a backward-error statement. Symplecticity creates a modified Hamiltonian, analyticity makes the defect in its numerical conservation exponentially small, and comparison of the modified and original Hamiltonians confines $\mathcal{K}$ to an $O(h^p)$ band for exponentially long times. Symmetry improves the parity of the modified expansion but is not, by itself, the source of the energy bound.

For the project, this theorem applies directly to the exact-root idealizations of Gauss–Legendre and the fully extended implicit ABBA methods. A literal implementation-level theorem additionally requires a regularity statement for the periodic spline, conditioning bounds for the implicit equations, and explicit control of nonlinear and round-off errors.

References

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